

Classificatie Pipeline - Patty Sausele 1146363

```
import numpy as np
import pandas as pd
import re
import joblib
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.metrics import f1_score
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from textblob import TextBlob

df = pd.read_csv('studentset.csv')
df
```

	transcript	label
0	So when I was a kid ... this was my team.(Laug...	1
1	This is how war starts. One day you're living ...	0
2	In the next 18 minutes, I'm going to take you ...	1
3	Before March, 2011, I was a photographic retou...	0
4	Chris Anderson: And now we go live to Caracas ...	1
...	...	...
1845	Most of the time, art and science stare at eac...	0
1846	For emotions, we should not move quickly to th...	0
1847	I published this article in the New York Times...	1
1848	I'll never forget the sound of laughing with m...	1
1849	I'm not at all a cook. So don't fear, this is ...	0

[1850 rows x 2 columns]

Data voorbereiden

Data opschonen:

```
def opschonen_tekst(transcript):
    transcript = re.sub(r'\\([^)]*\\)', '', transcript)
    transcript = transcript.lower()
    transcript = re.sub(r'^[a-z\s]', '', transcript)

    return transcript
```

Ik haal de speciale tekens en nummers uit de transcripts. Daarbij haal ik ook de woorden tussen haakjes (zoals laughter, music, applause) eruit, omdat deze niet relevant zijn.

```

df['transcript_schoon'] = df['transcript'].apply(opschonen_tekst)
df[['transcript', 'transcript_schoon']].head()

      transcript \
0  So when I was a kid ... this was my team.(Laug...
1  This is how war starts. One day you're living ...
2  In the next 18 minutes, I'm going to take you ...
3  Before March, 2011, I was a photographic retou...
4  Chris Anderson: And now we go live to Caracas ...

      transcript_schoon
0  so when i was a kid this was my teami stunk a...
1  this is how war starts one day youre living yo...
2  in the next minutes im going to take you on a...
3  before march i was a photographic retoucher b...
4  chris anderson and now we go live to caracas t...

features = df['transcript_schoon']
label = df['label']

X_train, X_test, y_train, y_test = train_test_split(features, label,
test_size=0.2, random_state=42)

```

Bag of Words language model:

```

count_vectorizer =
CountVectorizer(stop_words='english',min_df=200,max_df=0.95,binary=True)

X_train_bow = count_vectorizer.fit_transform(X_train)
X_test_bow = count_vectorizer.transform(X_test)

print(X_train_bow.shape)
print(X_test_bow.shape)

(1480, 632)
(370, 632)

def sentiment_score_berekenen(transcript):
    return TextBlob(transcript).sentiment.polarity

train_sentiment_scores = np.array([sentiment_score_berekenen(text) for
text in X_train])
test_sentiment_scores = np.array([sentiment_score_berekenen(text) for
text in X_test])

train_sentiment_scores = train_sentiment_scores.reshape(-1, 1)
test_sentiment_scores = test_sentiment_scores.reshape(-1, 1)

```

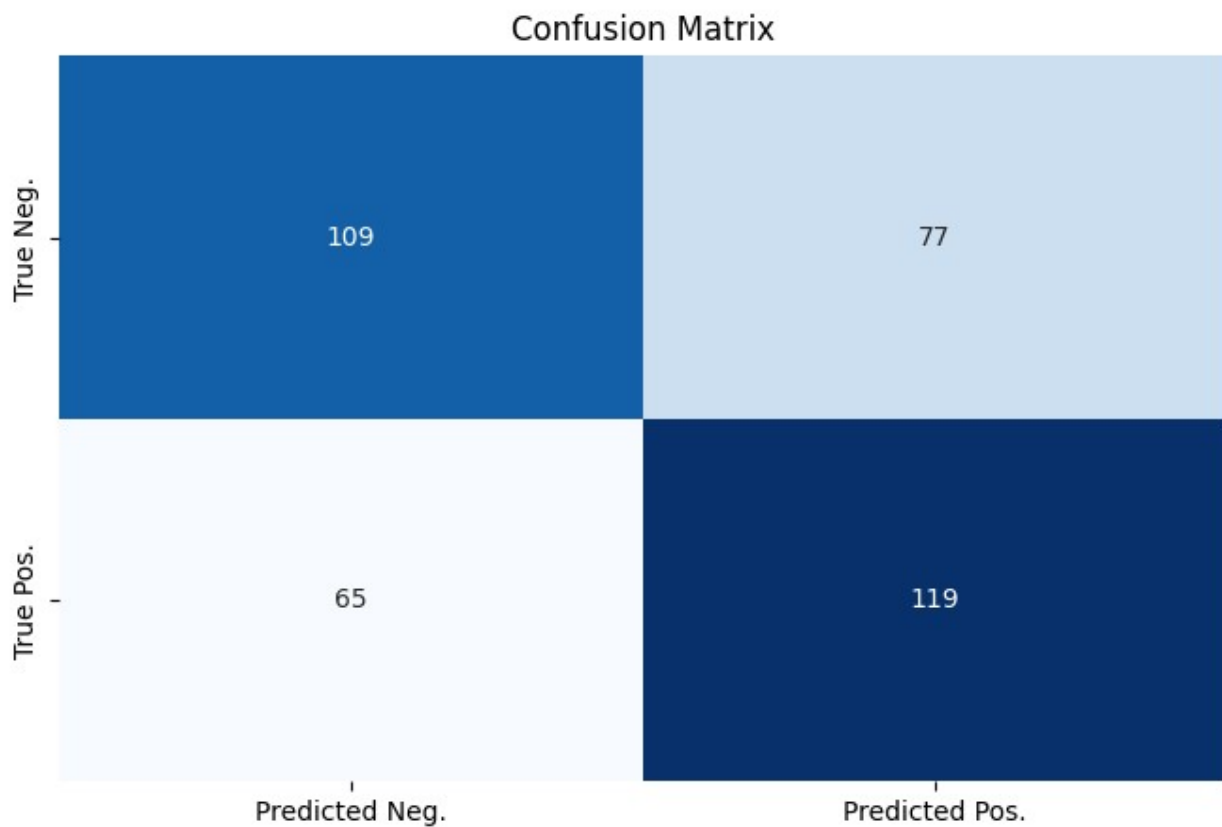


```

Pos.'],
                                index=['True Neg.', 'True Pos.'])

plt.figure(figsize=(8, 5))
sns.heatmap(cm_df, annot=True, fmt='d', cmap='Blues', cbar=False)
plt.title('Confusion Matrix')
plt.show()

```



Opslaan modellen

```

joblib.dump(classificatie_model, 'classificatie_model.pkl')
['classificatie_model.pkl']
joblib.dump(count_vectorizer, 'nlp_model.pkl')
['nlp_model.pkl']

```